

Artifact	Cells	Episodes/Cell	Total Episodes
Core grid (5 variants $\times$ 2 budgets $\times$ 2 seeds)	20	3	60
Where-bits block (tail50 vs head50, bA seed1)	2	3	6
Prior pilot FP16 context (m3_main2exp_v2)	1	10	10

Table 1: Inventory of completed artifacts consolidated in this paper.

WHERE BITS MATTER IN DINO-WM PLANNING: A CONSOLIDATED ICLR-STYLE REPORT FROM COM- PLETED M3 EXPERIMENTS

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Paper under double-blind review

ABSTRACT

This paper consolidates all completed quantization experiments currently available in our M3 pipeline for DINO-WM planning on Wall. The central question is whether low-bit planning quality depends only on bitwidth, or also on where precision is allocated. We report results from a strict run with 22 completed cells (66 evaluated episodes): a 20-cell core grid over FP16, uniform/mixed INT8, and uniform/mixed INT4 across two budgets and two seeds, plus a two-cell encoder localization block (tail vs head FP16 retention at 4-bit). We also include one earlier pilot FP16 artifact as context. The evidence is clear that 8-bit variants are robust and 4-bit variants are fragile. However, mixed INT4 is not uniformly superior to uniform INT4 across both budgets in this completed run; its advantage is budget-dependent. This paper provides the finalized tables/plots for the completed data, explains each visualization, and identifies the next experiments needed to convert budget-dependent signals into a stronger generalized claim.

1 INTRODUCTION

Recent world-model planners derive control from latent dynamics and representation quality. At the same time, low-bit quantization is increasingly used for deployment under strict memory and compute constraints. Our working hypothesis is that at low precision, module-aware allocation matters: preserving precision in representation-critical components may outperform uniform quantization at the same nominal bitwidth.

This report focuses on completed experiments only. It is not a speculative plan. Every number and figure is generated from saved outputs in the run directories.

2 EXPERIMENTAL PROTOCOL AND DATA CONSOLIDATION

Task and model. We evaluate DINO-WM on Wall using paired-goal planning episodes.

Hardware and backend. All consolidated runs here use Apple M3 with MPS backend.

Completed runs included in this paper. Table 1 clarifies what is and is not pooled. The strict run contributes 66 episodes total. The pilot FP16 row is reported separately as context because its protocol differs (10 episodes vs 3 per strict cell).

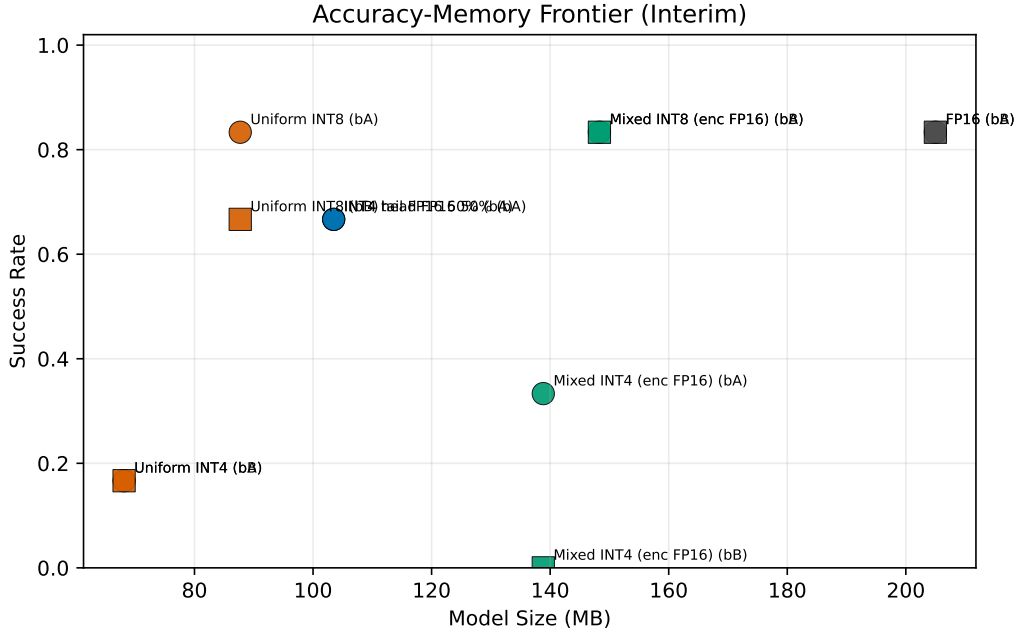


Figure 1: Accuracy-memory frontier from completed strict-run outputs. Marker color encodes variant family and annotations include budget ID.

Variant (strict run aggregate)	Mean Success	Mean Time (s)	Mean Size (MB)
FP16	0.8333	278.45	204.99
Uniform INT8	0.7500	301.33	87.72
Mixed INT8	0.8333	295.90	148.31
Uniform INT4	0.1667	294.38	68.12
Mixed INT4	0.1667	306.91	138.84

Table 2: Core strict-run aggregate over bA/bB and seeds 1/2.

3 MAIN RESULTS

3.1 OVERALL REGIME STRUCTURE: 8-BIT STABLE, 4-BIT FRAGILE

Figure 1 and Table 2 together show the first major pattern. The 8-bit variants remain strong and close to FP16 in success, while both 4-bit variants drop sharply. The frontier plot is useful because it shows this is not only an accuracy story; it also exposes memory compression tradeoffs directly. In this run, the smallest model (uniform INT4) is also the weakest among the shown families.

3.2 BUDGET-CONDITIONED BEHAVIOR AT 4-BIT

Figure 2 is the most important diagnostic for the core claim. It shows that mixed INT4 is better than uniform INT4 on bA, but worse on bB. Figure 3 and Table 3 make this explicit as signed deltas. So the completed data does *not* support a universal statement that mixed INT4 dominates uniform INT4 in this setup. Instead, the effect is budget-conditioned.

3.3 PAIRED STATISTICS AND WHERE-BITS LOCALIZATION

Paired statistics refine the mean-level view. Figure 4 shows: (1) bA mixed-vs-uniform INT4 delta is positive with CI touching zero, (2) bB delta is negative with CI touching zero, (3) tail50-vs-head50 delta is exactly zero in this sample. The confidence intervals are wide because paired units are small (6 for mixed-vs-uniform per budget, 3 for tail-vs-head).

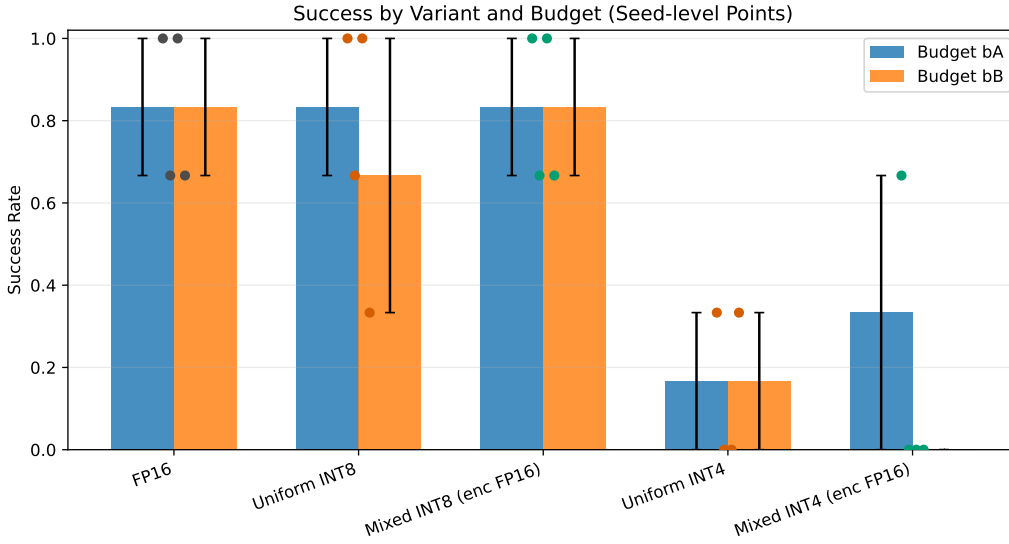


Figure 2: Budget-wise success bars with seed-level points for key variants.

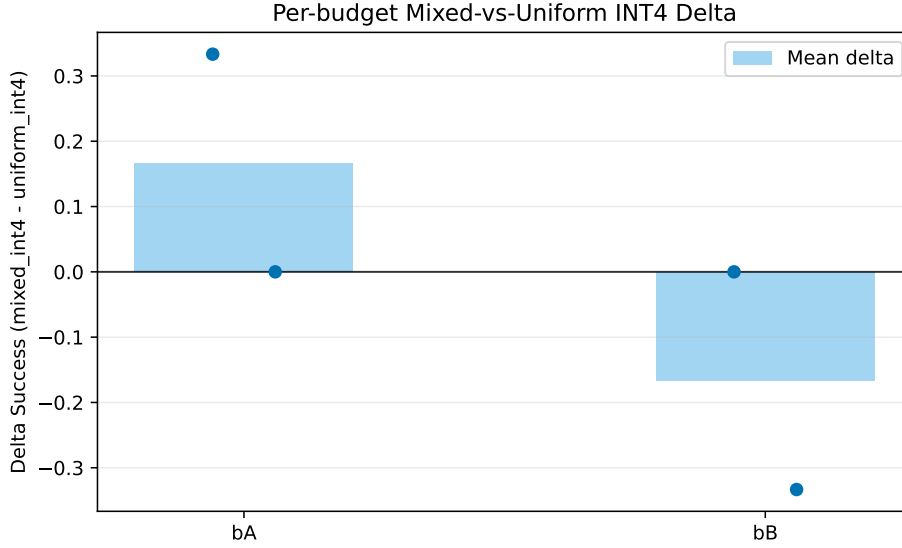


Figure 3: Budget-level delta $\Delta_{\text{success}} = \text{mixed INT4} - \text{uniform INT4}$ from strict-run means.

Figure 5 helps interpret the full matrix quickly. It confirms strong 8-bit performance, low 4-bit performance, and near-identical tail/head localization scores in the one-cell where-bits block (both 0.6667 on bA seed1). This localization result should be treated as provisional because it comes from a single seed and single budget.

4 WHAT THIS CONSOLIDATED EVIDENCE SUPPORTS

From completed outputs, we can support the following claims:

- 8-bit quantization (uniform/mixed) remains close to FP16 in this Wall setup.
- Moving to 4-bit substantially reduces success in aggregate.

Budget	Uniform INT4	Mixed INT4	Δ (Mixed – Uniform)
bA	0.1667	0.3333	+0.1667
bB	0.1667	0.0000	-0.1667

Table 3: Budget-conditioned 4-bit comparison over seeds 1 and 2.

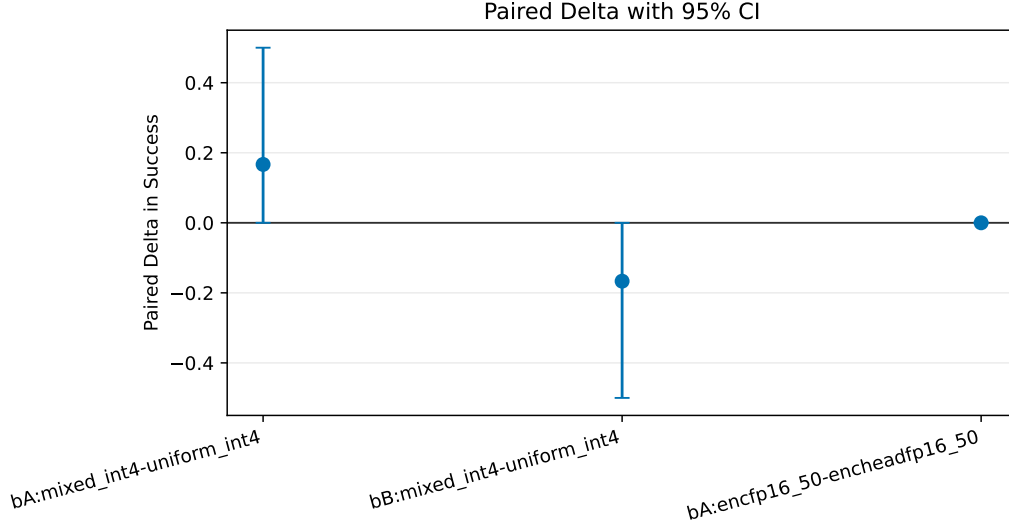


Figure 4: Paired deltas with 95% CIs from episode-level paired analysis JSON outputs.

- Mixed INT4 vs uniform INT4 is budget-sensitive rather than consistently positive in this completed run.
- The current tail-vs-head localization check is inconclusive at present sample size.

5 LIMITATIONS

The strict run uses small per-cell episode count (3), one environment (Wall), one checkpoint family, and limited where-bits coverage. These choices were intentional for runtime control, but they widen uncertainty. As a result, this paper should be interpreted as a consolidated completed-run report, not a final generalized theorem about all planning regimes.

6 CONCLUSION

This ICLR-style consolidated report combines all completed artifacts to date and presents a consistent story: bitwidth alone is not sufficient to explain behavior near the low-bit frontier, but current evidence is not yet enough to claim universal mixed-INT4 superiority. The data instead points to budget-conditioned effects, motivating the next phase: higher episode counts and broader budget sweeps to determine where mixed allocation reliably wins.

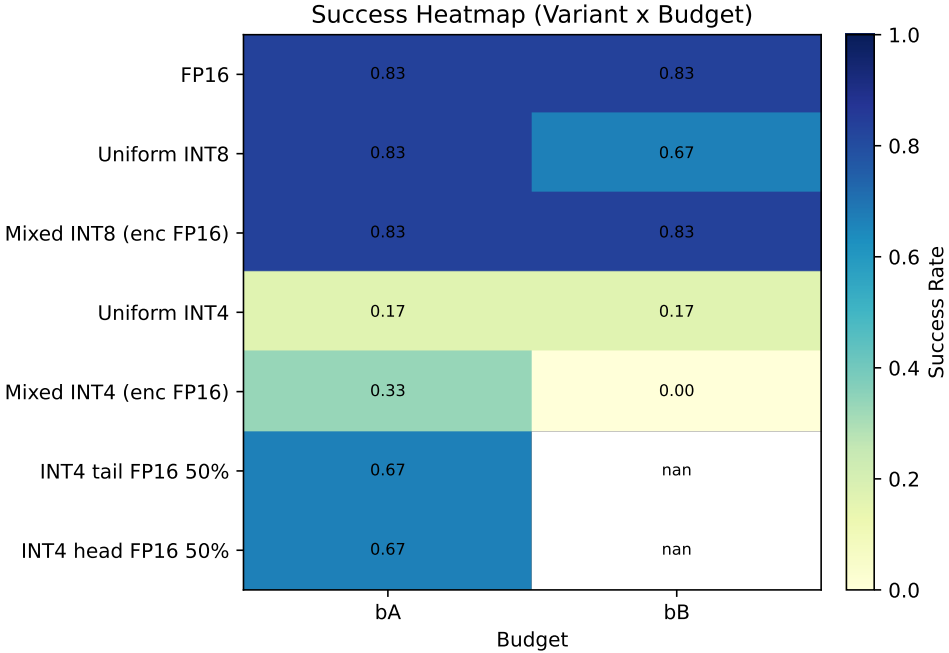


Figure 5: Variant-by-budget success heatmap, including where-bits variants (tail50 and head50).